

Speech and Language Processing Project Report

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Abstract

This report studies an adaptation of CRAB (Contrastive Representation and Multimodal Aligned Bottleneck) from speech emotion recognition to audio deepfake detection. The original speech-text architecture is reformulated as a bona fide/spoof classifier by combining self-supervised speech representations with a rhythm branch extracted from forced-alignment-based duration features. Experiments on ASVspoof 5 compare audio-only, audio-rhythm, and audio-text variants with different encoders and contrastive supervision settings. The results show that the audio-only Transformer provides the strongest baseline in this report, while rhythm fusion and MPCL require further refinement.

Keywords: audio deepfake detection, ASVspoof, self-supervised speech representation, contrastive learning, CRAB

I INTRODUCTION

Recent neural text-to-speech (TTS) and voice conversion (VC) systems can generate highly natural and speaker-similar speech [1]. As a result, audio deepfake detection (ADD), which aims to distinguish bona fide speech from spoofed or synthetic speech, has become an important problem in speech processing and information security. Reliable ADD systems are especially important because synthetic speech can be used to imitate a speaker's identity, bypass voice-based authentication, or spread misleading audio content.

In recent years, self-supervised learning (SSL) models have become important front-end components for speech-related tasks because they can learn rich representations from large amounts of unlabeled data. Models such as wav2vec 2.0 [2], HuBERT [3], and WavLM [4] have shown strong performance across various downstream tasks, including automatic speech recognition, speaker verification, emotion recognition, and spoofing detection. For ADD, these SSL representations provide strong acoustic cues, but synthetic speech may also contain unnatural temporal patterns that are not fully described by frame-level spectral rep-

resentations alone. For example, the duration of syllables, vowels, and consonants, as well as the rhythm between neighboring speech units, may reveal differences between human speech production and generated speech [5–7].

This report studies whether such rhythm information can be incorporated into the CRAB framework. The original CRAB model was designed for speech emotion recognition (SER), where speech and text representations are aligned through cross-modal attention and supervised at multiple intermediate layers using Multi Layer Contrastive Supervision (MLCS). In this work, we adapt the same representation-learning idea to ADD by replacing the original text branch with a rhythm branch extracted from forced-alignment outputs. The goal is to make the adapted architecture and its ablation behavior clear: the experiments compare audio-only, audio-rhythm, and audio-text variants, different sequence encoders, and the effect of contrastive supervision.

The main objectives and contributions of this report are summarized as follows:

1. We adapt the CRAB framework from speech emotion recognition (SER) to audio deepfake detection (ADD), reformulating the original emotion classification task as bona fide/spoof classification.
2. We construct a syllable-level rhythm branch from forced-alignment outputs, where each utterance is represented by duration-based tokens derived from syllable, vowel, and consonant intervals.
3. We describe an audio-rhythm multimodal architecture that encodes SSL speech representations and rhythm representations separately, then aligns them through cross-modal attention.
4. We conduct ablation experiments over modality choice, encoder type, and multi-layer contrastive supervision to clarify the role of each component in the adapted detection system.

The rest of this report is organized as follows. Section 2 reviews related work on prosody-based ADD and the original CRAB model. Section 3 describes the pro-

posed methodology, including prosody preprocessing, the rhythm encoder, and the audio-rhythm adaptation of CRAB for ADD. Section 4 presents the experimental setup for the SER reproduction and ASVspoof 5 ADD experiments. Section 5 reports the SER reproduction results, ADD ablation results, and analysis. Section 6 concludes the report.

II RELATED WORK

2.1 Prosody Features for Audio Deepfake Detection

Early prosody-based methods mainly rely on hand-crafted or explicitly extracted prosodic descriptors. These approaches commonly use F_0 -based statistics, jitter, shimmer, harmonics-to-noise ratio (HNR), rhythm, pitch, accent-related cues, and phoneme-duration information to characterize unnatural prosodic patterns in synthetic speech [5–7]. In particular, Pitch Imperfect [5] evaluates acoustic-prosodic deepfake detection with mean and standard deviation of F_0 , jitter, shimmer, and mean and standard deviation of HNR. Other feature-fusion methods combine prosody with speaker verification embeddings, pronunciation features, and wav2vec representations to improve synthetic speech detection and cross-dataset generalization [6, 7].

More recent work incorporates prosody directly into representation learning and training objectives. HuLA [8] uses F_0 prediction and voiced/unvoiced classification as auxiliary tasks so that a self-supervised speech backbone can learn natural prosodic variation, improving robustness to expressive, emotional, and cross-lingual synthetic speech. ProSDD [9] further models speaker-conditioned prosodic variation through pitch, voice activity, and energy, and learns transferable prosodic representations via supervised masked prediction.

2.2 CRAB model

The CRAB model was first proposed for speech emotion recognition (SER), where speech and text inputs are jointly modeled by a bimodal Cross-Modal Transformer [10]. Its architecture uses WavLM to extract frame-level speech representations and RoBERTa [11] to extract token-level text representations. After modality-specific projection and recurrent encoding, cross-modal attention aligns the two streams, and attention pooling converts them into utterance-level

representations for emotion classification.

The central training idea in CRAB is Multi Layer Contrastive Supervision (MLCS), which adds contrastive supervision to several intermediate representations instead of applying supervision only at the final classifier. Each supervised layer is connected to a Contrastive Supervision Leg (CSL), a small projection head that maps the intermediate representation into a contrastive embedding space. The CSL is trained with Multi Positive Contrastive Learning (MPCL), where samples with the same emotion label are treated as positives and samples with different labels are treated as negatives. For an anchor embedding a and candidate embeddings $\{b_i\}_{i=1}^K$, MPCL first defines a similarity distribution

$$q_i = \frac{\exp(a \cdot b_i / \tau)}{\sum_{j=1}^K \exp(a \cdot b_j / \tau)}, \quad (1)$$

where τ is a temperature parameter and embeddings are ℓ_2 -normalized. The target distribution assigns probability mass only to candidates that share the same label as the anchor:

$$c_i = \frac{\mathbf{1}_{\text{match}}(a, b_i)}{\sum_{j=1}^K \mathbf{1}_{\text{match}}(a, b_j)}. \quad (2)$$

The MPCL loss is the cross-entropy between these two distributions,

$$\mathcal{L}_{\text{MPCL}} = - \sum_{i=1}^K c_i \log q_i. \quad (3)$$

Compared with ordinary supervised contrastive learning, this formulation explicitly represents all same-class candidates as multiple positives in a categorical target distribution. In MLCS, the final SER objective combines the standard classification loss with the average MPCL loss from all selected CSLs:

$$\mathcal{L}_{\text{ser}} = \mathcal{L}_{\text{CE}}(m_K(x), y) + \alpha \frac{1}{|\Omega|} \sum_{i \in \Omega} \mathcal{L}_{\text{MPCL}}(m_i(x), y), \quad (4)$$

where Ω denotes the set of layers with CSLs, and α controls the strength of contrastive supervision. This design encourages modality-specific, cross-modal, and fused representations to become discriminative before the final prediction layer, which is the part of CRAB most relevant to our adaptation for audio deepfake detection.

Table I: Summary of ASVspoof 5 database [12] and TextGrid generated after preprocessing.

Subset	Utterances		TextGrid		Missing rate (%)
	Bona fide	Spoofed	Bona fide	Spoofed	
Train	18,797	163,560	18,754	163,003	0.33
Dev	31,334	109,616	30,858	97,413	8.99
Eval-T1	138,688	542,086	136,044	504,640	5.89

III METHODOLOGY

3.1 Prosody Feature Preprocessing

To obtain reliable temporal units for prosodic feature extraction, we first preprocess each utterance through a three-stage pipeline consisting of transcription, forced alignment, and syllable boundary generation. Figure 1 illustrates the overall process flow-chart of prosody feature preprocessing. In the transcription stage, WhisperX [13] is applied to the raw audio signals to generate utterance-level transcripts. These transcripts serve as the textual input for the subsequent alignment procedure.

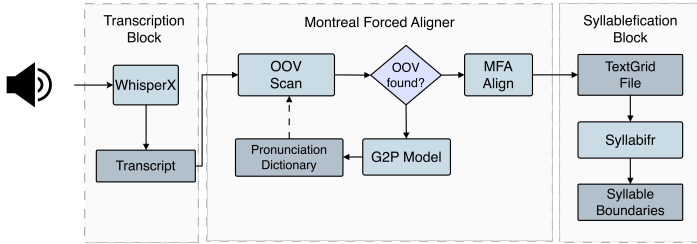


Figure 1: Overview of the preprocessing pipeline.

Given the audio-transcript pairs, we then employ the Montreal Forced Aligner (MFA) to obtain fine-grained temporal alignments [14].

Before alignment, the generated transcripts are scanned for out-of-vocabulary (OOV) words with respect to the pronunciation dictionary. If OOV entries are detected, a grapheme-to-phoneme (G2P) model is used to predict their pronunciations, and the pronunciation dictionary is updated accordingly. This OOV detection and dictionary expansion process is repeated until all transcript tokens can be covered by the dictionary.

After the dictionary covers all transcript tokens, MFA is used to align the speech signal with the normalized transcript, producing a TextGrid file that contains word- and phone-level timing information.

Finally, the TextGrid output is passed to our syllabification module, which converts the phone-level align-

ment into syllable-level boundaries. These syllable boundaries provide the temporal anchors required for extracting prosodic descriptors, such as duration-related features and syllable-level rhythmic patterns. Through this preprocessing procedure, each ASVspoof5 utterance is transformed into a structured representation with transcript, phonetic alignment, and syllable timing information for downstream prosody analysis.

3.2 Prosody Encoder

Following the rhythm encoder design of Liu and Lu [15], we encode prosodic timing as a sequence of duration tokens. The rhythm encoder represents each valid syllable position using three interval types:

$$\mathcal{S} = \{\text{SYL}, \text{V}, \text{CONS}\}, \quad (5)$$

where SYL denotes the whole syllable interval, V denotes the vocalic interval, and CONS denotes the consonant interval. For interval type $s \in \mathcal{S}$, $\text{Int}_{s,i}$ denotes the duration of the i -th interval in an utterance with N valid syllable positions.

For each interval type, we compute three durational measurements. The first is the raw duration of the interval, $\text{Int}_{s,i}$. The second is the deviation of the current interval from the utterance-level mean duration of the same interval type:

$$\text{Devi}\mu_{s,i} = \text{Int}_{s,i} - \mu_{\text{Int}_{s,i}} \quad (6)$$

The third is the local durational difference between consecutive intervals, defined as

$$\mu\text{Diff}_{s,i} = \frac{\text{Int}_{s,i} - \text{Int}_{s,i+1}}{(\text{Int}_{s,i} + \text{Int}_{s,i+1})/2} \quad (7)$$

At the n -th interval, $\mu\text{Diff}_{s,n}$ is replaced by nPVI-Int, i.e., the normalized pairwise variability index for a specific type of speech interval, which measures the speech rhythm of an utterance:

$$\begin{aligned} \mu\text{Diff}_{s,n} &= \text{nPVI-Int} \\ &= \frac{\sum_{k=1}^{n-1} \left| \frac{\text{Int}_k - \text{Int}_{k+1}}{(\text{Int}_k + \text{Int}_{k+1})/2} \right|}{n-1}. \end{aligned} \quad (8)$$

The resulting duration sequence is denoted as

$$D = [d_1, d_2, \dots, d_n] \in \mathbb{R}^{9 \times n}, \quad (9)$$

where each duration token $d_i \in \mathbb{R}^9$ contains nine features that capture durational variability. Each token is

first linearly projected into a d_{model} -dimensional representation using a learnable projection weight $W_d \in \mathbb{R}^{d_{model} \times 9}$. To preserve the temporal order of the sequence, sinusoidal positional encoding is then added to the projected duration embeddings, following the original Transformer [16]. Finally, the input embedding of the rhythm encoder is defined as

$$e_i = LN(W_d \cdot d_i + p_i), \quad (10)$$

where $LN(\cdot)$ denotes layer normalization, and $p_i \in \mathbb{R}^{d_{model}}$ is the positional encoding vector at position i .

3.3 CRAB for ADD

Figure 2 shows the proposed adaptation of CRAB for audio deepfake detection. The original CRAB model is designed for SER and uses a speech-text Cross-Modal Transformer with WavLM and RoBERTa [11] in the original CRAB design [10]. In this work, we keep the main representation learning principle of CRAB, but reformulate the architecture for ADD. The task label is changed from an emotion category to a binary deepfake label $y \in \{0, 1\}$, where 0 denotes bona fide speech and 1 denotes spoofed speech. More importantly, the text branch is replaced by a rhythm branch. Although transcript or contextual information can be useful in some context-dependent ADD settings, lexical content is not a consistently necessary or reliable cue for spoof detection. In contrast, synthetic speech often exhibits acoustic and prosodic artifacts in duration, syllable timing, pitch, and vowel-consonant rhythm patterns. The resulting model is an audio-rhythm CRAB network.

Given an input waveform x , a self-supervised speech encoder first extracts a frame-level acoustic representation S :

$$S = f_{ssl}(x), \quad S \in \mathbb{R}^{F \times D_s}. \quad (11)$$

Here, $f_{ssl}(\cdot)$ denotes the SSL speech encoder, F is the number of speech frames, and D_s is the speech feature dimension. In this report, the SSL encoder can be instantiated with wav2vec 2.0/XLS-R style representations [2]. This branch provides the model with frame-level acoustic information from the waveform.

In parallel, the duration sequence D produced by the prosody encoder is converted into a rhythm representation R :

$$R = f_{rhythm}(D), \quad R \in \mathbb{R}^{L \times D_r}. \quad (12)$$

Here, $f_{rhythm}(\cdot)$ denotes the rhythm encoder, L is the number of rhythm tokens, and D_r is the rhythm feature dimension. This branch summarizes syllable-level timing information, so it describes how speech units are distributed over time rather than what words are spoken.

Because the speech and rhythm representations have different feature dimensions, they are projected into a shared hidden space before fusion:

$$\tilde{S} = LN(W_s \cdot S), \quad \tilde{R} = LN(W_r \cdot R). \quad (13)$$

In this equation, W_s and W_r are modality-specific projection matrices, and $LN(\cdot)$ denotes layer normalization. The projected sequences $\tilde{S}$ and $\tilde{R}$ therefore have compatible hidden dimensions, which allows the later cross-modal attention block to compare the two streams.

After projection, each modality is encoded by its own bidirectional GRU:

$$H_s = \text{BiGRU}_s(\tilde{S}), \quad H_r = \text{BiGRU}_r(\tilde{R}). \quad (14)$$

The resulting representations satisfy $H_s \in \mathbb{R}^{F \times 2h}$ and $H_r \in \mathbb{R}^{L \times 2h}$, where h is the hidden size of one GRU direction. This step lets each branch model temporal context within its own sequence before the two modalities interact.

The cross-modal block then aligns the two streams through multi-head attention. The speech stream attends to the rhythm stream as follows:

$$A_s = \text{MHA}(Q = H_s, K = H_r, V = H_r). \quad (15)$$

In this attention operation, speech features are used as queries, while rhythm features are used as keys and values. The output A_s can be interpreted as rhythm-aware speech information.

The reverse direction is computed in the same way, where rhythm features attend to the speech stream:

$$A_r = \text{MHA}(Q = H_r, K = H_s, V = H_s). \quad (16)$$

Here, rhythm features are used as queries, and speech features are used as keys and values. The output A_r can be interpreted as speech-aware rhythm information.

The attention outputs are then added back to their original unimodal representations through residual connections:

$$Z_s = H_s + A_s, \quad Z_r = H_r + A_r. \quad (17)$$

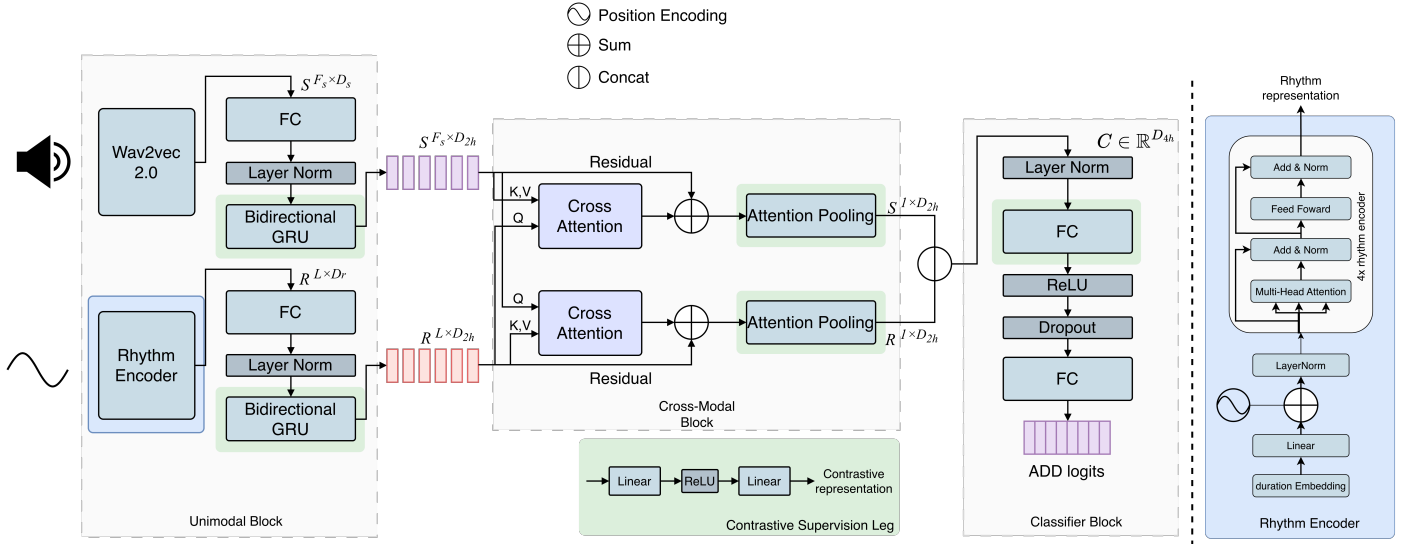


Figure 2: Overall architecture of the proposed audio-rhythm CRAB model.

The residual design keeps the original speech and rhythm information while injecting cross-modal context. As a result, Z_s is the speech representation after rhythm alignment, and Z_r is the rhythm representation after speech alignment.

Since the final prediction is made at the utterance level, attention pooling is used to summarize each aligned sequence into one fixed-length vector:

$$z_s = \text{AttPool}(Z_s), \quad z_r = \text{AttPool}(Z_r). \quad (18)$$

Here, z_s and z_r are utterance-level speech and rhythm vectors, respectively. Attention pooling allows the model to assign larger weights to frames or rhythm tokens that are more informative for spoof detection.

The two utterance-level vectors are concatenated to form the final multimodal representation:

$$C = [z_s; z_r] \in \mathbb{R}^{4h}. \quad (19)$$

The vector C contains information from both the acoustic stream and the rhythm stream. It is then passed through the classifier block, which consists of layer normalization, a fully connected layer, ReLU activation, dropout, and a final fully connected layer that outputs two logits for bona fide and spoof classes.

We also adapt Multi Layer Contrastive Supervision (MLCS) from the original CRAB model to the ADD setting. Each green-highlighted module in Figure 2 is connected to a Contrastive Supervision Leg (CSL). For the i -th supervised layer, the CSL maps an intermediate representation $m_i(x)$ into a contrastive representation u_i :

$$u_i = g_i(m_i(x)). \quad (20)$$

In this equation, $g_i(\cdot)$ is the projection head of the i -th CSL. For an anchor representation u_i , other samples in the mini-batch with the same ADD label are treated as positive samples; that is, bona fide samples are positives for a bona fide anchor, and spoofed samples are positives for a spoofed anchor. Samples with the opposite bona fide/spoof label are treated as negative samples. Following the MPCL formulation in Equations (1) and (2), the similarity distribution is computed over all candidate samples, while the target distribution assigns probability mass only to the positive samples. Therefore, MLCS encourages modality-specific, aligned, and fused representations to become label-discriminative before the final classifier.

The overall training objective combines the ordinary classification loss with the averaged contrastive loss from all selected CSLs:

$$\mathcal{L}_{\text{add}} = \mathcal{L}_{\text{CE}}(o, y) + \alpha \frac{1}{|\Omega|} \sum_{i \in \Omega} \mathcal{L}_{\text{MPCL}}(u_i, y). \quad (21)$$

Here, o denotes the classifier logits, y is the bona fide/spoof label, Ω is the set of layers with CSLs, and α controls the strength of contrastive supervision. During inference, the CSL branches are removed, and prediction only uses the audio-rhythm fusion path and the final classifier. This keeps inference simple while using MLCS during training to improve the separability of bona fide and spoof representations.

IV Experimental Setup

4.1 IEMOCAP

We use IEMOCAP [17] to reproduce the SER setting of the original CRAB model before adapting the framework to ADD. IEMOCAP contains two-speaker conversations collected from five sessions, with utterance-level emotion annotations and corresponding transcripts.

The IEMOCAP data collection mainly targets five emotion categories: neutral, happiness, excitement, anger, and sadness. Following the common four-class SER protocol, happiness and excitement are merged into the same positive emotion class. The final label set is therefore {neutral, happy/excited, angry, sad}. The dataset is split by speaker/session to avoid direct speaker leakage: Session 5 is used as the test set, one actor from Session 4 is used as the development set, and the remaining utterances are used for training.

For the reproduced SER experiment, WavLM Large [4] is used as the speech encoder and RoBERTa Base [11] is used as the text encoder. The model is trained for 20 epochs with AdamW and a CosineAnnealingLR scheduler. We set the learning rate of the SSL encoder and text encoder to 1×10^{-6} , and the learning rate of the SER classification head to 1×10^{-5} . The batch size is set to 4 with gradient accumulation over 8 steps. The training objective combines weighted cross-entropy with MPCL, and the best checkpoint is selected according to development-set UAR. We report both WAR and UAR on the test set.

4.2 ASVspoof 5 Dataset

We use ASVspoof 5 [12] as a benchmark to evaluate the performance of our model. A common practical preprocessing choice is to use a fixed-length strategy, where each utterance is normalized to a 10-second segment by cropping longer audio and repeating or padding shorter audio. Although this setting is practical for batch-based training, it may discard spoofing cues located outside the selected segment. In contrast, we process the entire utterance without fixed-duration cropping, allowing the model to exploit the full temporal information available in the ASVspoof 5 recordings.

Table I summarizes the ASVspoof 5 preprocessing results in this study. Models are trained on the training set and validated on the validation set to select the best

checkpoint for testing.

Our proposed CRAB for ADD models is trained with a learning rate of 1×10^{-5} using the AdamW optimizer, without any data augmentation. The number of training epochs is set to 20, the batch size is set to 8, and gradient accumulation is performed over 8 steps.

V RESULTS

We first reproduce the SER setting of the CRAB model on IEMOCAP [17] and MSP-Podcast 2.0 [18]. Table II presents a comparative evaluation of all baseline systems and the proposed CRAB model. The bold values indicate the best performance within each group, underlined values denote the second-best results, and green highlighting is used to emphasize the proposed model or components introduced by our approach.

Table II: Comparison of SER performance on IEMOCAP and MSP-Podcast 2.0.

Model	IEMOCAP		MSP-Podcast 2.0	
	WAR	UAR	WAR	UAR
WavLM Baseline [19]	0.6616	0.6832	0.5131	0.4105
FocalSer [20]	0.6676	0.6943	0.2892	0.3360
MemoCMT [21]	0.6964	0.7123	0.6239	0.3285
Medusa [22]	0.6605	0.6778	0.4157	0.3549
CRAB [10]	<u>0.7348</u>	<u>0.7585</u>	<u>0.5232</u>	<u>0.4390</u>
CRAB (re-implement)	0.7365	<u>0.74292</u>	0.4997	<u>0.4194</u>

Table II shows that the re-implemented CRAB model obtains competitive performance on IEMOCAP, with a WAR of 0.7365 and a UAR of 0.7429. Although the reproduction does not exactly match the reported CRAB results, it shows a comparable trend and confirms that the main architecture is usable as the basis for ADD adaptation.

Table III reports ADD performance using EER and min-DCF. The audio-only Transformer without MPCL gives the lowest numerical result, with an EER of 4.81% and a min-DCF of 0.1358. However, this result is unexpectedly strong and should be interpreted cautiously, since the current implementation may still contain coding or evaluation issues. Possible sources include data split handling, label mapping, score aggregation, evaluation-script usage, checkpoint loading, and padding or masking behavior. The audio-only GRU improves from 8.34% to 5.39% EER with MPCL, but this trend should also be verified after the code path is checked.

Table III: Comparison of audio deepfake detection performance on ASVspoof5 [12].

Model	Modality	Encoder	MPCL	ASVspoof5	
				EER(%)	min-DCF
AASIST [23]	Audio	RawNet-2	—	6.08	0.1645
Nes2Net [24]	Audio	—	—	6.13	0.1568
ProSDD [9]	Audio+Prosody	—	—	7.38	—
AudioRhythmModel	Audio	GRU	×	8.34	0.2394
AudioRhythmModel	Audio	GRU	✓	<u>5.39</u>	<u>0.156</u>
AudioRhythmModel	Audio	Transformer	×	4.81	0.1358
AudioRhythmModel	Audio	Transformer	✓	6.44	0.1792
AudioRhythmModel	Audio+Rhythm	GRU	×	12.33	0.3364
AudioRhythmModel	Audio+Rhythm	GRU	✓	7.56	0.1989
AudioRhythmModel	Audio+Rhythm	Transformer	×	8.16	0.2148
AudioRhythmModel	Audio+Rhythm	Transformer	✓	5.66	0.1641
AudioTextModel	Audio+Text	GRU	×	7.03	0.2012
AudioTextModel	Audio+Text	GRU	✓	7.26	0.2096
AudioTextModel	Audio+Text	Transformer	×	6.17	0.1774
AudioTextModel	Audio+Text	Transformer	✓	6.94	0.3559

The multimodal results are weaker than the audio-only Transformer result, but this gap should not be taken as evidence that rhythm information is ineffective. The rhythm branch has a weaker front end because it uses a Transformer encoder trained from scratch, while the audio branch uses a pretrained SSL checkpoint. Moreover, the rhythm pipeline depends on forced alignment, syllable boundaries, padding, masking, normalization, and fusion, so small implementation errors can easily affect the result. Therefore, the current ADD table should be treated as a preliminary ablation for debugging rather than a final comparison.

VI CONCLUSION

This report adapted the CRAB framework from speech emotion recognition to audio deepfake detection by reformulating the task as bona fide/spoof classification and replacing the original text branch with a rhythm branch. The adapted model combines SSL speech representations, syllable-level duration features, cross-modal attention, attention pooling, and Multi Layer Contrastive Supervision to study audio-rhythm fusion for ADD.

The ablation results should be interpreted as preliminary because the unusually low audio-only Transformer EER may indicate coding or evaluation issues. Before drawing conclusions about rhythm fusion or MPCL, future work should first verify the data split, label mapping, scoring pipeline, checkpoint loading, masking, and multimodal code path. After the implementation is verified, stronger or pretrained rhythm encoders can be evaluated more fairly.

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